Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18MEDE735]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

TECHNOLOGY OF SURFACE COATING

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Explain the following terms with suitable examples: **i.** Corrosion **ii.** Sand blasting **iii.** Shot peening **iv.** Hydro blasting	**08**
	b.	Explain the role of surface engineering in the development of sustainable products with suitable examples.	**06**
2.	**a.**	Summarize the need for surface pretreatments in coating industry	**05**
	b.	Explain the process of hot dipping.	**05**
	c.	Describe the process of anodizing with suitable illustrations	**04**
3.	**a.**	Describe the process of Physical Vapor Deposition (PVD) with suitable illustrations. Also list the major advantages and disadvantages of the process	**08**
	b.	Summarize the importance of surface modifications using directed energy beam approach over other coating methods	**06**
		OR	
4.	**a.**	Discuss the process of Chemical Vapor Deposition (CVD) with suitable illustrations. Highlight the major applications of CVD.	**08**
	b.	Explain the following terms with suitable illustrations: **i.** Sputtering **ii.** Plasma spray method **iii.** Atmospheric oxidation	**06**
5.	**a.**	Elucidate the process of carburizing. Also list the major applications employing carburizing	**07**
	b.	Differentiate the process of carbonitriding and chromizing	**07**
		OR	
6.	**a.**	Highlight the major applications of various high temperature coating technique.	**07**
	b.	Review the process of siliconizing and list the major applications employing the method	**07**

7.	**a.**	Compare the chemical and plasma coating approaches for industrial applications from environmental perspective.	**07**
	b.	Propose the application of any economical coating technique for a machining industry.	**07**

OR

8.	**a.**	Propose a suitable coating methodology for the application in Naval industries.	**07**
	b.	Examine the major factors affecting the process of anodizing. Also comment on controlling these factors to achieve effective coating.	**07**

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